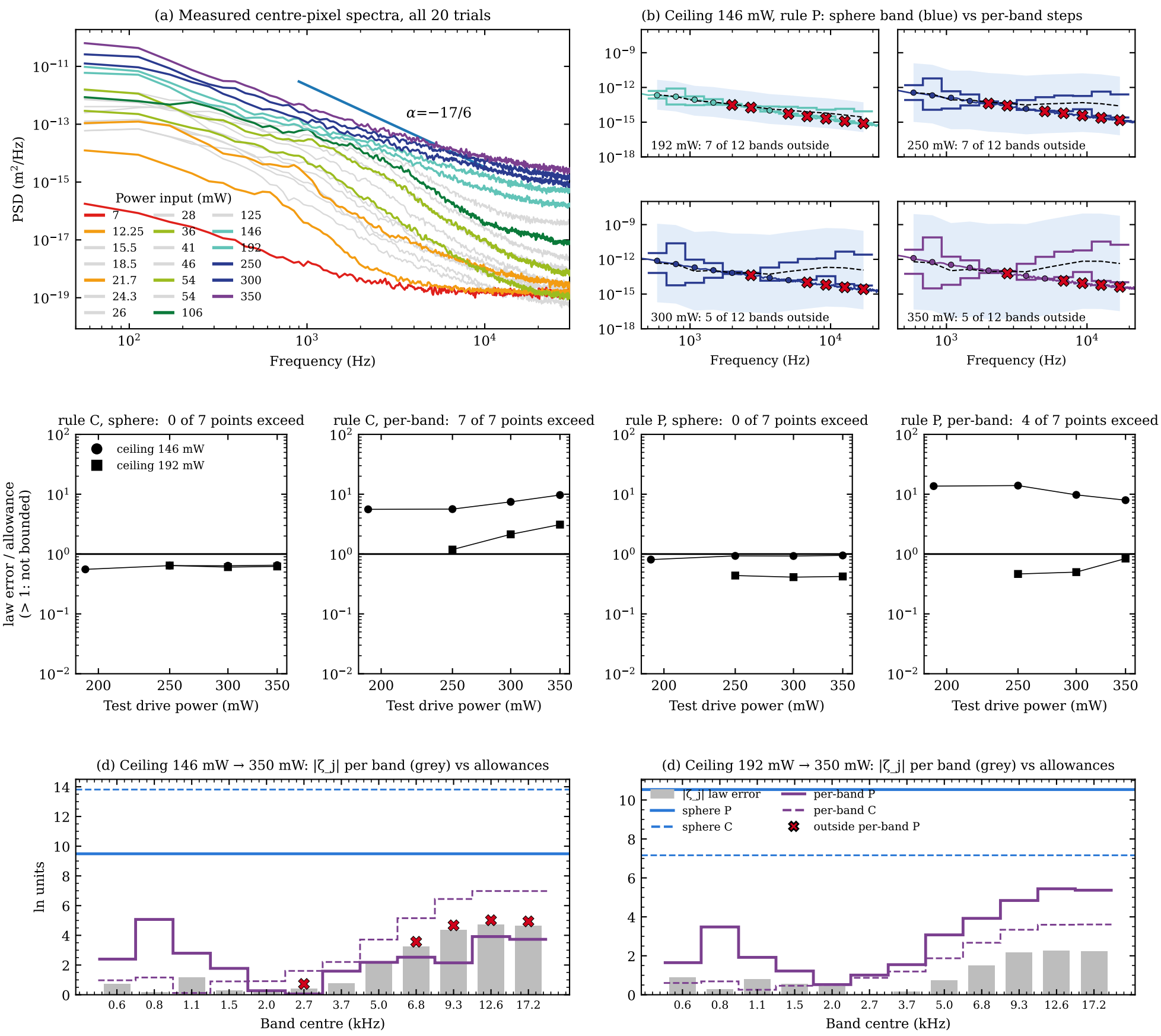
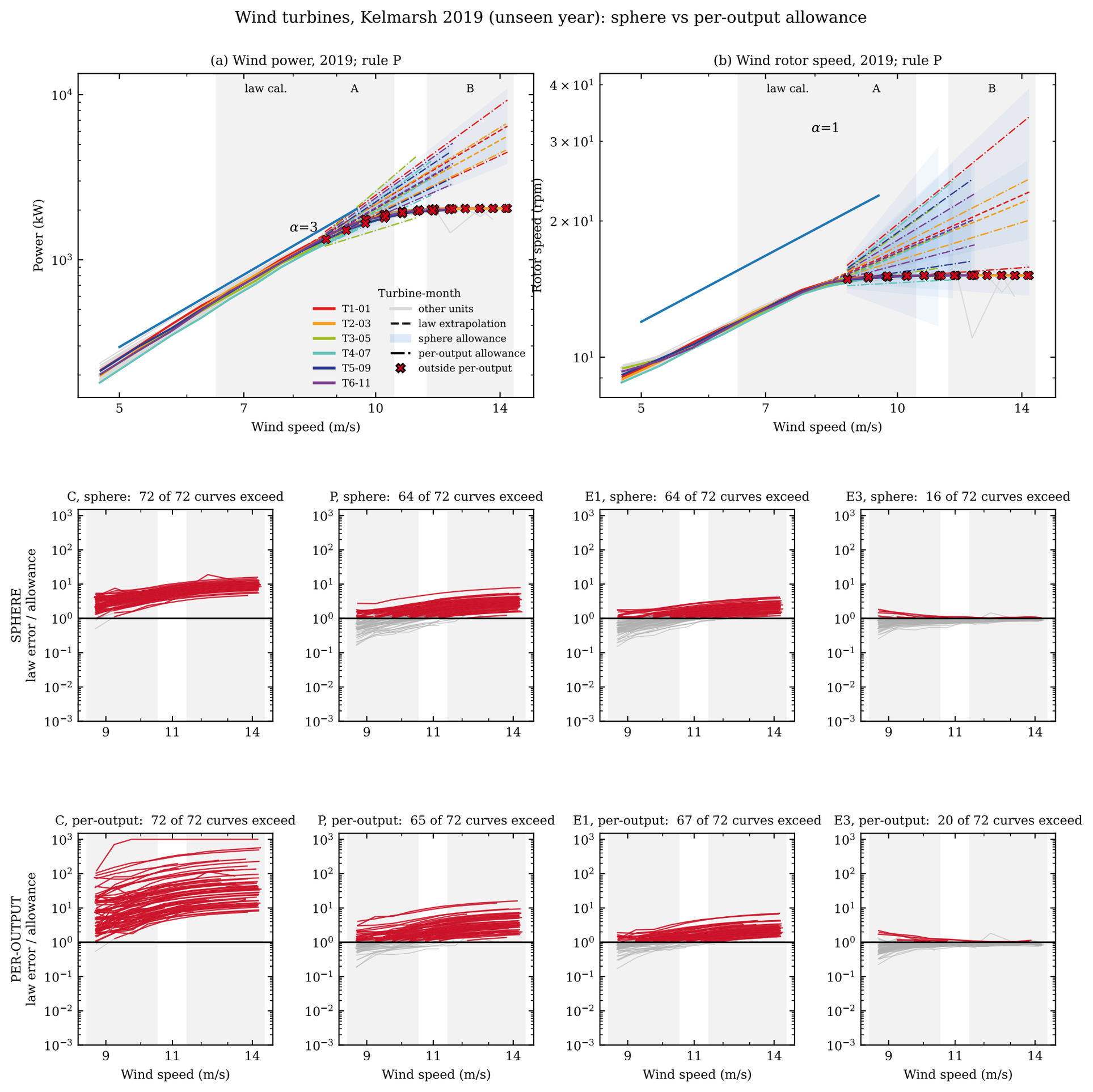

Capillary waves: sphere vs per-band allowance
(a) Measured centre-pixel spectra, all 20 trials
PSD (m²/Hz)
Frequency (Hz)
α=−17/6
Power input (mW)
7
12.25
15.5
18.5
21.7
24.3
26
28
36
41
46
54
54
106
125
146
192
250
300
350
(b) Ceiling 146 mW, rule P: sphere band (blue) vs per-band steps
192 mW: 7 of 12 bands outside
250 mW: 7 of 12 bands outside
300 mW: 5 of 12 bands outside
350 mW: 5 of 12 bands outside
Frequency (Hz)
rule C, sphere: 0 of 7 points exceed
rule C, per-band: 7 of 7 points exceed
rule P, sphere: 0 of 7 points exceed
rule P, per-band: 4 of 7 points exceed
ceiling 146 mW
ceiling 192 mW
law error / allowance
(> 1: not bounded)
Test drive power (mW)
(d) Ceiling 146 mW → 350 mW: |ζ_j| per band (grey) vs allowances
(d) Ceiling 192 mW → 350 mW: |ζ_j| per band (grey) vs allowances
ln units
|ζ_j| law error
sphere P
sphere C
per-band P
per-band C
outside per-band P
Band centre (kHz)


Wind turbines, Kelmarsh 2019 (unseen year): sphere vs per-output allowance
(a) Wind power, 2019; rule P
(b) Wind rotor speed, 2019; rule P
law cal.
A
B
α=3
α=1
Power (kW)
Rotor speed (rpm)
Wind speed (m/s)
Turbine-month
T1-01
T2-03
T3-05
T4-07
T5-09
T6-11
other units
law extrapolation
sphere allowance
per-output allowance
outside per-output
C, sphere: 72 of 72 curves exceed
P, sphere: 64 of 72 curves exceed
E1, sphere: 64 of 72 curves exceed
E3, sphere: 16 of 72 curves exceed
C, per-output: 72 of 72 curves exceed
P, per-output: 65 of 72 curves exceed
E1, per-output: 67 of 72 curves exceed
E3, per-output: 20 of 72 curves exceed
SPHERE
law error / allowance
PER-OUTPUT
law error / allowance
Wind speed (m/s)